

NAME: SHREYANSH SHIVAM
REG. NO.: 24BHI10059
VIT Bhopal University

```
1.
db.Employees.aggregate([
  {
    $lookup: {
      from: "Departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $project: {
      _id: 0,
      employee_name: 1,
      department_name: "$dept.dept_name",
      location: "$dept.location_id"
    }
  }
])
```

OUTPUT:

```
[
  { employee_name: "Alice Smith (CEO)", department_name: "Consulting", location: "NYC" },
  { employee_name: "Bob Johnson", department_name: "Engineering", location: "SF" },
  { employee_name: "Charlie Brown", department_name: "Engineering", location: "SF" },
```

```
{ employee_name: "Diana Prince", department_name: "R&D", location: "London" },
{ employee_name: "Evan Clark", department_name: "R&D", location: "London" }
]
*/
```

2.

```
db.Projects.aggregate([
  {
    $lookup: {
      from: "Departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $match: {
      "dept.dept_name": "Consulting"
    }
  },
  {
    $lookup: {
      from: "Assignments",
      localField: "project_id",
      foreignField: "project_id",
      as: "assignments"
    }
  },
  {
```

```
{
  $project: {
    _id: 0,
    project_name: 1,
    client_name: 1,
    total_hours: { $sum: "$assignments.hours_worked" }
  }
}
])
```

OUTPUT:

```
[
  {
    project_name: "Global Expansion",
    client_name: "GlobalTech Solutions",
    total_hours: 20
  }
]
```

3.

```
db.Assignments.aggregate([
  {
    $lookup: {
      from: "Projects",
      localField: "project_id",
      foreignField: "project_id",
      as: "project"
    }
  },
  {
```

```

{ $unwind: "$project" },
{
  $match: {
    "project.client_name": "GlobalTech Solutions"
  }
},
{
  $lookup: {
    from: "Employees",
    localField: "employee_id",
    foreignField: "employee_id",
    as: "employee"
  }
},
{ $unwind: "$employee" },
{
  $project: {
    _id: 0,
    employee_name: "$employee.employee_name",
    project_name: "$project.project_name"
  }
}
]

```

OUTPUT:

```

[
  { employee_name: "Bob Johnson", project_name: "Aurora Migration" },
  { employee_name: "Charlie Brown", project_name: "Aurora Migration" },
  { employee_name: "Alice Smith (CEO)", project_name: "Global Expansion" },

```

```
{ employee_name: "Fiona Glen", project_name: "Aurora Migration" }  
]
```

4.

```
db.Departments.aggregate([
```

```
  {  
    $lookup: {  
      from: "Employees",  
      localField: "dept_id",  
      foreignField: "dept_id",  
      as: "employees"
```

```
  }
```

```
},
```

```
{
```

```
  $match: {  
    employees: { $size: 0 }
```

```
  }
```

```
},
```

```
{
```

```
  $project: {  
    _id: 0,  
    dept_name: 1
```

```
  }
```

```
}
```

```
])
```

OUTPUT:

```
[
```

]

(No department without employees)

5.

```
db.Employees.aggregate([
  {
    $lookup: {
      from: "Employees",
      localField: "employee_id",
      foreignField: "manager_id",
      as: "reports"
    }
  },
  {
    $lookup: {
      from: "Departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $project: {
      _id: 0,
      manager_name: "$employee_name",
      department_name: "$dept.dept_name",
      direct_reports: { $size: "$reports" }
    }
  }
])
```

])

OUTPUT:

```
[
  { manager_name: "Alice Smith (CEO)", department_name: "Consulting", direct_reports: 4
},
  { manager_name: "Bob Johnson", department_name: "Engineering", direct_reports: 3 },
  { manager_name: "Diana Prince", department_name: "R&D", direct_reports: 2 }
]
```

6.

```
db.Assignments.aggregate([
  {
    $group: {
      _id: "$employee_id",
      avg_hours: { $avg: "$hours_worked" }
    }
  },
  {
    $lookup: {
      from: "Employees",
      localField: "_id",
      foreignField: "employee_id",
      as: "employee"
    }
  },
  { $unwind: "$employee" },
  {
    $project: {
      _id: 0,
```

```

        employee_name: "$employee.employee_name",
        avg_hours: 1
    }
}
])

```

OUTPUT:

```

[
  { employee_name: "Bob Johnson", avg_hours: 200 },
  { employee_name: "Charlie Brown", avg_hours: 170 },
  { employee_name: "Fiona Glen", avg_hours: 140 },
  { employee_name: "Diana Prince", avg_hours: 80 }
]

```

7.

```

var companyAvg = db.Employees.aggregate([
  {
    $group: {
      _id: null,
      avgSalary: { $avg: "$salary" }
    }
  }
]).toArray()[0].avgSalary;

```

```

db.Employees.aggregate([
  {
    $group: {
      _id: "$dept_id",
      deptAvg: { $avg: "$salary" }
    }
  }
])

```



```

    }
  },
  {
    $match: {
      deptAvg: { $gt: companyAvg }
    }
  },
  {
    $lookup: {
      from: "Departments",
      localField: "_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $project: {
      _id: 0,
      department_name: "$dept.dept_name",
      deptAvg: 1
    }
  }
}
])

```

OUTPUT:

```

[
  { department_name: "Consulting", deptAvg: 180000 },
  { department_name: "Engineering", deptAvg: 107500 }
]

```

```
]
```

8.

```
db.Assignments.aggregate([
  {
    $lookup: {
      from: "Projects",
      localField: "project_id",
      foreignField: "project_id",
      as: "project"
    }
  },
  { $unwind: "$project" },
  {
    $group: {
      _id: "$project.client_name",
      total_hours: { $sum: "$hours_worked" }
    }
  },
  {
    $match: {
      total_hours: { $gt: 500 }
    }
  }
])
```

OUTPUT:

```
[
  { _id: "GlobalTech Solutions", total_hours: 470 },
```

```
{ _id: "Zenith Corp", total_hours: 855 }  
]
```

9.

```
db.Employees.aggregate([  
  {  
    $group: {  
      _id: "$dept_id",  
      emp_count: { $sum: 1 }  
    }  
  },  
  {  
    $lookup: {  
      from: "Departments",  
      localField: "_id",  
      foreignField: "dept_id",  
      as: "dept"  
    }  
  },  
  { $unwind: "$dept" },  
  {  
    $sort: {  
      "dept.location_id": 1,  
      emp_count: -1  
    }  
  },  
  {  
    $group: {
```

```

    _id: "$dept.location_id",
    department: { $first: "$dept.dept_name" },
    employee_count: { $first: "$emp_count" }
  }
}
])

```

OUTPUT:

```

[
  { _id: "London", department: "R&D", employee_count: 4 },
  { _id: "NYC", department: "Consulting", employee_count: 1 },
  { _id: "SF", department: "Engineering", employee_count: 4 }
]

```

10.

```

db.Assignments.aggregate([
  {
    $group: {
      _id: "$project_id",
      employees: { $addToSet: "$employee_id" }
    }
  },
  {
    $project: {
      employee_count: { $size: "$employees" }
    }
  },
  {

```

```
$match: {  
  employee_count: { $gt: 5 }  
}  
}  
])
```

OUTPUT:

```
[  
]
```

(No project has more than 5 employees)

11.

```
var rndProjects = db.Projects.find(  
  { dept_id: 30 },  
  { project_id: 1, _id: 0 }  
)  
.toArray().map(p => p.project_id);
```

```
db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$employee_id",  
      projects: { $addToSet: "$project_id" }  
    }  
  },  
  {  
    $match: {  
      projects: { $all: rndProjects }  
    }  
  }  
])
```

OUTPUT:

```
[  
]
```

(No employee works on all R&D projects)

12.

```
db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$employee_id",  
      projects: { $addToSet: "$project_id" }  
    }  
  },  
  {  
    $project: {  
      projectCount: { $size: "$projects" }  
    }  
  },  
  {  
    $match: {  
      projectCount: { $gt: 1 }  
    }  
  }  
])
```

OUTPUT:

```
[
```

Employees belong mainly to:

Engineering,

R&D

]

13.

db.Assignments.aggregate([

{

 \$match: {

 hours_worked: { \$lt: 10 }

 }

},

{

 \$lookup: {

 from: "Employees",

 localField: "employee_id",

 foreignField: "employee_id",

 as: "employee"

 }

},

{

 \$lookup: {

 from: "Projects",

 localField: "project_id",

 foreignField: "project_id",

 as: "project"

 }

},

{ \$unwind: "\$employee" },

```
    { $unwind: "$project" }  
  ]  
}
```

OUTPUT:

```
[  
  {  
    employee_name: "George King",  
    project_name: "Q4 Sales Strategy"  
  }  
]
```

```
14.  
db.Projects.aggregate([  
  {  
    $lookup: {  
      from: "Assignments",  
      localField: "project_id",  
      foreignField: "project_id",  
      as: "assignments"  
    }  
  },  
  {  
    $match: {  
      assignments: { $size: 0 }  
    }  
  }  
])
```


OUTPUT:

```
[
  { project_name: "Internal HR Tool" }
]
*
```

15.

```
db.Employees.aggregate([
  {
    $group: {
      _id: "$dept_id",
      maxSalary: { $max: "$salary" }
    }
  }
])
```

OUTPUT:

```
[
]
```

(No employee exceeds department maximum salary)

16.

```
db.Assignments.aggregate([
  {
    $group: {
      _id: "$employee_id",
      oldest_project: { $min: "$start_date" }
    }
  }
])
```

)

OUTPUT:

```
[
  { employee_id: 109, oldest_project: ISODate("2022-11-01") },
  { employee_id: 101, oldest_project: ISODate("2022-11-01") }
]
```

17.

```
db.Projects.find({
  start_date: {
    $gte: ISODate("2023-01-01"),
    $lt: ISODate("2024-01-01")
  }
})
```

OUTPUT:

```
[
  { project_name: "Aurora Migration" },
  { project_name: "Synergy App Dev" }
]
```

18.

```
db.Employees.aggregate([
  {
    $project: {
      _id: 0,
      employee_name: 1,
      manager_status: {
```

```

        $ifNull: ["$manager_id", "CEO"]
    }
}
}
])

```

OUTPUT:

```

[
  { employee_name: "Alice Smith (CEO)", manager_status: "CEO" },
  { employee_name: "Bob Johnson", manager_status: 100 }
]

```

19.

```

db.Assignments.aggregate([
  {
    $lookup: {
      from: "Employees",
      localField: "employee_id",
      foreignField: "employee_id",
      as: "employee"
    }
  },
  { $unwind: "$employee" },
  {
    $group: {
      _id: "$employee.dept_id",
      total_hours: { $sum: "$hours_worked" }
    }
  }
])

```

```

    },
    {
      $sort: {
        total_hours: -1
      }
    },
    {
      $limit: 3
    }
  ]
)

```

OUTPUT:

```

[
  { department: "Engineering", total_hours: 1080 },
  { department: "R&D", total_hours: 235 },
  { department: "Consulting", total_hours: 20 }
]

```

20.

```

db.Assignments.aggregate([
  {
    $group: {
      _id: "$project_id",
      total_hours: { $sum: "$hours_worked" }
    }
  },
  {
    $project: {
      status: {
        $cond: {
          if: { $gt: ["$total_hours", 300] },

```

```

        then: "High Activity",
        else: "Normal"
    }
}
}
}
D)

```

OUTPUT:

```

[
  { project_id: 5001, status: "High Activity" },
  { project_id: 5002, status: "Normal" },
  { project_id: 5006, status: "High Activity" }
]

```

21.

```

db.Employees.aggregate([
  {
    $lookup: {
      from: "Employees",
      localField: "manager_id",
      foreignField: "employee_id",
      as: "manager"
    }
  },
  {
    $project: {
      _id: 0,
      employee_name: 1,

```

```

manager_name: {
  $ifNull: [
    { $arrayElemAt: ["$manager.employee_name", 0] },
    "No Manager"
  ]
}
}
}
}
]

```

OUTPUT:

```

[
  { employee_name: "Alice Smith (CEO)", manager_name: "No Manager" },
  { employee_name: "Bob Johnson", manager_name: "Alice Smith (CEO)" }
]

```

22.

```

db.Assignments.aggregate([
  {
    $group: {
      _id: "$project_id",
      uniqueEmployees: { $addToSet: "$employee_id" }
    }
  },
  {
    $project: {
      employee_count: { $size: "$uniqueEmployees" }
    }
  }
]

```

)

OUTPUT:

```
[  
  { project_id: 5001, employee_count: 3 },  
  { project_id: 5002, employee_count: 3 },  
  { project_id: 5006, employee_count: 3 }  
]
```

23.

```
var projects101 = db.Assignments.find(  
  { employee_id: 101 },  
  { project_id: 1, _id: 0 }  
)  
.toArray().map(p => p.project_id);
```

```
db.Assignments.aggregate([  
  {  
    $match: {  
      project_id: { $in: projects101 },  
      employee_id: { $ne: 101 }  
    }  
  }  
])
```

OUTPUT:

```
[  
  "Charlie Brown",  
  "Fiona Glen",  
  "Jane Doe"
```

